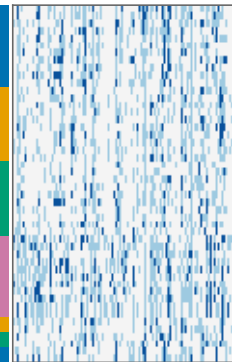


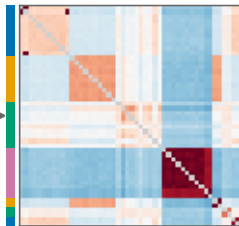
# aarobust-kin: robust PCA of the GRM with a kinship threshold (default for small $N$ )

genotypes  $G$



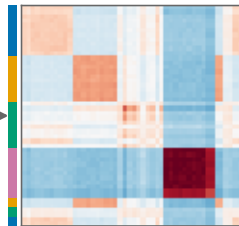
standardise

GRM  $C$ , diag. unobserved



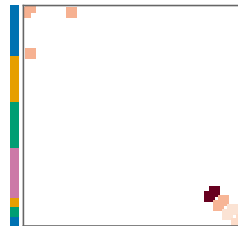
PCP  
 $C = L + S$

$L$ : completed structure



+

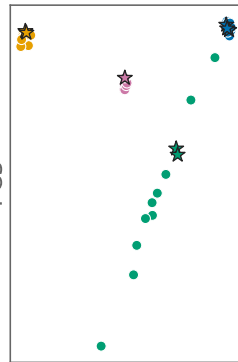
$S$ : related pairs



eigen-  
vectors

PC3

PCs of  $L$



● CEU ● CHB ● MXL ● YRI ★ relative